

```
pip install nltk
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.13/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.13/dist-packages (from nltk) (8.5.0)
Requirement already satisfied: joblib in /usr/local/lib/python3.13/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex<=2021.8.3 in /usr/local/lib/python3.13/dist-packages (from nltk) (2025.
Requirement already satisfied: tqdm in /usr/local/lib/python3.13/dist-packages (from nltk) (4.67.3)
```

```
import nltk
nltk.download('wordnet')
```

```
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Package wordnet is already up-to-date!
True
```

```
from nltk.stem import PorterStemmer, SnowballStemmer, WordNetLemmatizer
ps, ss, lem = PorterStemmer(), SnowballStemmer('english'), WordNetLemmatizer()
words3= ['running', 'studies', 'better', 'flies', 'meeting', 'disappointing']
print(f"{'word':<15}{'Porter':<12}{'Snowball':<12}{'Lemma (v)':<12}")
for w in words3:
    print(f"{'w':<15}{'ps.stem(w)':<12}{'ss.stem(w)':<12}{'lem.lemmatize(w,pos='v')':<12}")
```

word	Porter	Snowball	Lemma (v)
running	run	run	run
studies	studi	studi	study
better	better	better	better
flies	fli	fli	fly
meeting	meet	meet	meet
disappointing	disappoint	disappoint	disappoint

```
import nltk
```

```
nltk.download('punkt')
nltk.download('stopwords')
```

```
import nltk, warnings
warnings.filterwarnings('ignore')
```

```
for file in ['punkt.tab', 'stopwords', 'wordnet', 'movie_reviews.txt']:
```

```
for pkg in [ punkt_tag , stopwords , wordnet , movie_reviews ]:
    nltk.download(pkg,quiet=True)
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Package stopwords is already up-to-date!
```

```
corpus=["My Name is Muskan.",
        "I am a @Data, Science. Student.",
        "I am currently in 5th Semester.",
        "I am pursuing. my Data Science degree from K.R. Mangalam University.",
        "My University is based in Sohna, Gurugram.",]
with open('corpus.txt', 'w') as f:
    f.write('\n'.join(corpus))
print('corpus.txt written:', len(corpus), 'documents')
docs=open('corpus.txt').read().splitlines()
```

```
corpus.txt written: 5 documents
```

```
from nltk.tokenize import word_tokenize, sent_tokenize
from nltk.corpus import stopwords
import string
stop= set(stopwords.words('english'))
toks= word_tokenize(' '.join(docs).lower())
clean=[t for t in toks if t not in stop and t not in string.punctuation]
print('Before:', len(toks), 'After:', len(clean), 'tokens')
print('First 15 cleaned tokens: ', clean[:15])
```

```
Before: 45 After: 19 tokens
First 15 cleaned tokens: ['name', 'muskan', 'data', 'science', 'student', 'currently', '5th', 'semester', '']
```

```
import re
def clean_text(t):
    t=re.sub(r'http\S+', '', t)
    t=re.sub(r'@\w+', '',t)
    t=re.sub(r'\d+', '', t)
```

```

c=re.sub(r'\s+', ' ',c)
t=re.sub(r'^A-Za-z\s',' ',t)
return re.sub(r'\s+', ' ',t).strip()
print('Before:', docs[1])
print('After:', clean_text(docs[1]))

```

Before: I am a @Data, Science. Student.
 After: I am a Science Student

```

%matplotlib inline
from nltk import FreqDist
import matplotlib.pyplot as plt
fd=FreqDist(clean)
print('10 most common words:')
for w, cnt in fd.most_common(10):
    print(f'{w:<12}{cnt}')
print('Hapaxes (first 8):', fd.hapaxes()[:8])
plt.figure(figsize=(8,4))
fd.plot(20, title='Top-20 Word Frequencies')

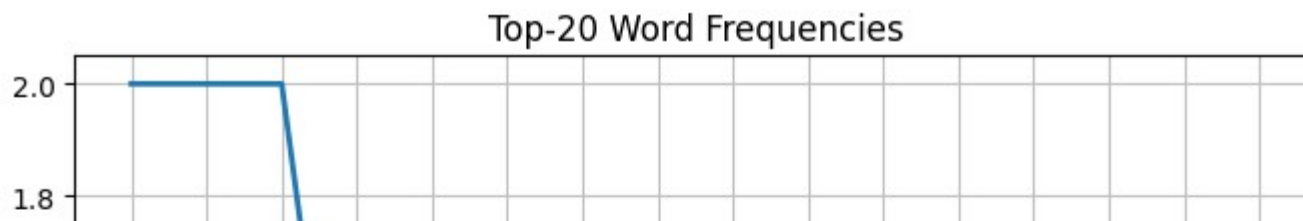
```

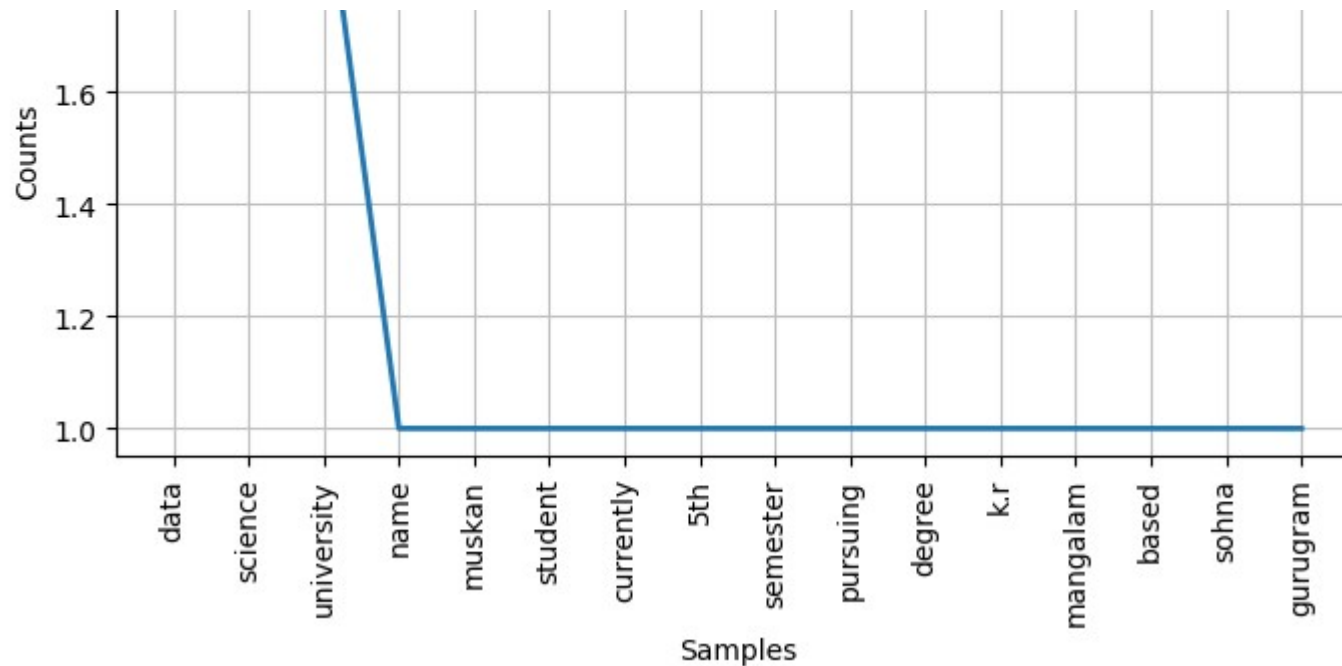
10 most common words:

data	2
science	2
university	2
name	1
muskan	1
student	1
currently	1
5th	1
semester	1
pursuing	1

Hapaxes (first 8): ['name', 'muskan', 'student', 'currently', '5th', 'semester', 'pursuing', 'degree']

<Axes: title={'center': 'Top-20 Word Frequencies'}, xlabel='Samples', ylabel='Counts'>





```
def preprocess(t):
    t=clean_text(t.lower())
    toks=word_tokenize(t)
    toks=[w for w in toks if w not in stop]
    return [lem.lemmatize(w,pos='v')for w in toks]

for d in docs[:2]:
    print(d,'\n->', preprocess(d),'\n')
total=sum(len(preprocess(d)) for d in docs)
print('Corpus after full pipeline:',total, 'clean tokens from', len(docs), 'documents')
```

```
My Name is Muskan.
-> ['name', 'muskan']
```

```
I am a @Data, Science. Student.
-> ['science', 'student']
```

```
Corpus after full pipeline: 19 clean tokens from 5 documents
```